Physical Account (D6, Annex C) -- every number we assumed, narrowed or overrode

Only ASSUMED or narrowed values are listed; Annex A/B values and Graziano's errata (G4 = 990 kg; Scenario B = G1+G3+G4+G5 = 9,560 kg) are used as given. Full list with provenance: ASSUMPTIONS.yaml.

Class	What we assumed	Value / range	Why	Where used in code
Electrical power	Reserve, trip, lines, peaks	200 kW cruise, 100 kW after a trip (no line lost); LINE_A_FWD (bay1, mast) / LINE_B_AFT (bay2, hangar, roof, deck, ramp), 150 kW each; peaks count as committed; G3 25 kW and G4 35 kW reserved before they draw	Brief gives no feeder topology; fail-safe	vessel_baseline.yaml:8-16, budgets.py:: evaluate_eight_class_budgets, ship_state.py::FirmReservedLoad, degradation.py:: execute_graceful_degradation
Heat rejection	Seawater, C(T), loop loss, inertia, air	24 C cruise, 30 C hot/degraded, stale feed -> 30 C; 160 kW below 12 C; loop fault x0.60 (72 kW at 30 C); +30 kW x 60 s once per 600 s, never for sustained loads; 1 kW = 1 kW heat; hangar roof air 2,000 W like the mast	Judge the voyage, not the 16 C pier	mission_envelope_a10.yaml, budgets.py:: compute_cooling_water_capacity_w
Positions	Bay sharing, mast, roof vs drone, ramp	bay1 = bay1_a + bay1_b; only mast_3 free; G5 antenna on the hangar roof only under ROOF_OUTSIDE_G4_RECOVERY_V1 (unverified -> D1 YES_IF, D3 refuses recovery); flight deck / ramp exclusive; ramp 5,500 kg with cradle (brief 5 t)	Graziano: declare and enforce the roof/drone rule	budgets.py:: evaluate_roof_drone_clearance, scheduling.py:: evaluate_roof_drone_clearance_safe, areas.py
Mass / stability	Multipliers, margin kept	x1.0 deck/bays, x1.2 hangar/roof, x3.0 mast; 10,000 kg and 12,000 kg-eq as hard lines, no hidden buffer (Scenario B keeps 440 kg / 2,147 kg-eq)	Brief limits, no extra margin	vessel_baseline.yaml:125-130, budgets.py:: evaluate_eight_class_budgets
Data links	SATCOM, EMCON, acoustic drop	SATCOM 8 nominal / 6 cruise / 4 hot / 2 degraded / floor 1 Mbit/s, 0 in EMCON; G3 HOLD handshake 15 s (10-60 s), no ACK by 60 s -> escalate; layered water, manoeuvre or deck machinery cut the link -> G3 holds itself after 120 s	Brief A.5: 2-8 Mbit/s, 10-60 s latency, link breaks	budgets.py, acoustic.py, acoustic_link_policy.yaml, replay.py
State transitions	Quiet, course, speed, deck per crew	quiet 180 s; course settle 90 s; recovery at 5 kn; clear deck 240 s (14) / 600 s (6) / impossible (0)	A.6 ranges, mid values	scheduling.py:: DEFAULT_TRANSITION_TIMING_POLICY
Compute, response time	Allocation, solver budget	4 nodes x 80 %, least-loaded node; exhaustive search (6 D3 candidates, all displacement sets), no timeout; every run checked against 30 / 2 / 5 / 60 s (measured: ms)	Small problem, exact answer	placement.py:: find_feasible_placement, admission.py, arbitration.py, degradation.py
Priorities, crew	Weights, confirmation, forbidden at 0	survey_primary / rescue_primary weights and shed order; D3 utility = survey continuity + 2.2 x landing margin; confirm 30 s (14) / 120 s (6) / none (0); at 0 no deck clearing, no G4 recovery, no physical install	Swappable file decides; safety is never weighed	data/priorities/*.yaml, arbitration.py:: _evaluate_candidate_feasibility_and_score, admission.py
Anything else	Guest behaviour, environment, invented	Airborne G4 / submerged G3 never shed; G5 split container/antenna; own radar cuts G1 to 70 % at sea unless QUIET_RADIO (>= 6 crew); one-genset-down window 100 kW at 30 C for Scenario A; Golden Gate SST 16.3 C (advisory); G6 card (420 kg) invented	Cards and brief silent	ASSUMPTIONS.yaml, degradation.py:: DEPLOYED_VEHICLE_MODES, capability_effects.py, admission.py:: evaluate_mission_envelope_a10

Scenario B ledger (actual / committed): 57.5 / 81.5 kW -> 82.5 / 106.5 kW at T=480 s (G4 35 kW reserved); genset trip at T=720 s: 106.5 > 100 kW -> G5 45 -> 8 kW -> 45.5 / 69.5 kW. Water 103.8 -> 67.5 kW committed (<= 72 kW even with loop loss at 30 C). Only two 4-guest sets pass mass; {G1,G2,G3,G4} = 10,035 kg fails.

What would change if the vessel had one more mast position (mast_4)?

Closing question: we would add one guest mast slot, mast_4 (<= 60 kg, <= 0.6 m2, 2,000 W air). Today G1 (1,500 W) and the G5 antenna (1,200 W) cannot share mast_3 (2,700 > 2,000 W), so the antenna sits on the hangar roof and every G4 recovery depends on a roof-clearance condition. With mast_4 the antenna leaves the roof (9,853 -> 9,898 kg-eq, 2,102 kg-eq margin left), the drone conflict disappears and D1/D3 lose a conditional rule. It does not fix mass: {G1,G2,G3,G4} (10,035 kg) still fails.